The food model: specification (version 4, M101)

Version 4 is a new model built from first principles. It replaces versions 1-3; their history is in the round files and in section 9. The reference implementation is `plan/tools/food_model.py`, which solves the model exactly. A worked example is in `food_model_worked_example.md`.

1. Principles

	Principle	What it fixes in the model
(P1)	**Conservation.** A city eats only what it holds.	consumption ≤ availability
(P2)	**Physiology bounds intake.** The stomach (fullness), energy turnover, the liver's capacity for protein and the tolerance for single foods set ceilings.	four constraints and a ceiling per good
(P3)	**Needs are absolute.** Energy follows work; nutrients and water have fixed daily needs.	the adequacy terms
(P4)	**Priority** (M88, M96). Nutrition first: being fed and covering needs. Then well-being. Nutrition and well-being are separate quantities, ordered, never added.	a lexicographic program
(P5)	**Harm is a consequence, not a motive.** People do not ration alcohol or tobacco by their harm; the harm is an outcome.	harm terms as outputs
(P6)	**The city as a whole** (M101). All quantities are per person of the city's population, averaged. Body reserves belong to the growth formula; seasons are below the turn length.	no classes, stores or seasons

2. Sets

Symbol	Meaning
$\mathcal{G}$, index g	food, drink and stimulant goods
$\mathcal{S} \subset \mathcal{G}$	staples: Wheat, Rice, Corn, Potato, Salt
$\mathcal{W} = \mathcal{G} \setminus \mathcal{S}$	goods that also give well-being: animal foods, fruit, sugar, olives, wine, beer, tea, coffee, tobacco
$\mathcal{D} \subset \mathcal{G}$	drinks: Dairy, Wine, Alcohol, Tea, Coffee
$\mathcal{T} \subset \mathcal{D}$	brewed stimulants without energy: Tea, Coffee
$\mathcal{F} \subset \mathcal{G}$	fruit: Apple, Banana, Lemon
$\mathcal{H} \subset \mathcal{G}$	harmful goods: Wine, Alcohol (alcohol); Tobacco

Symbol	Meaning
$\mathcal{I}$, index i	adequacy components (15): fullness, energy, digestible protein, vitamins C, A, B1, B3, folate, B12, iron, zinc, calcium, iodine, potassium, sodium

3. Parameters

3a. Per good (data files; one row per good)

Symbol	Meaning	Unit	Source
e_g	energy per gram as stored, eaten share included	kcal/g	nutrition.json (FAO)
$p_g,\ f_g$	protein, fat per gram	g/g	nutrition.json
d_g	protein quality (DIAAS, capped at 1)	–	nutrition.json
n_{gi}	nutrient i per gram, after bioavailability b_{gi}	mg or µg/g	USDA SR Legacy, MEXT; b_{gi}: iron 0.18/0.16 animal, 0.10/0.16 plant (FAO/WHO 2004); maize niacin 1/3 (Carpenter 1983); otherwise 1
σ_g	fullness per kcal (white bread = 1); drinks × 0.5	–	Holt et al. 1995 where measured; fitted formula otherwise (`satiety_fit.py`)
ω_g	water per gram of food eaten	ml/g	USDA SR Legacy
h_g	hydration index ($g \in \mathcal{D}$): hydration from a volume of the drink relative to the same volume of water	–	nutrition.json (milk 1.5)
u_g	ceiling per good: $\min(250\text{ g}/f_g,\ 3{,}000\text{ g milk},\ 116\text{ g lemon})$	g/(p·d)	the user (M86, M93); estimate (lemons)
α_g	harmful substance per gram ($g \in \mathcal{H}$): alcohol; tobacco	g/g	USDA SR Legacy; tobacco to add
δ_g	share of a year's leftover that spoils	–	nutrition.json (FAO storage)
$\ell_g \chi_g$	caffeine per gram ($g \in \mathcal{T}$)	mg/g	MEXT brewing: tea 43 L/kg × 200 mg/L, coffee 15 L/kg × 600 mg/L

3b. Scalars

Symbol	Meaning	Value	Source
B	basal metabolic rate of the average adult	1,536 kcal/(p·d)	FAO/WHO/UNU 2004; EFSA weights
Λ_k	activity level of work kind k (vigorous, active, sedentary)	2.25, 1.76, 1.53	FAO 2004, Table 5.1

Symbol	Meaning	Value	Source
$\Lambda^{\max}$	highest sustainable activity level	2.40	FAO 2004, Table 5.3
κ, γ	caffeine: extra expenditure at the dose	150 kcal at 600 mg	Dulloo et al. 1989
$\bar{\sigma}$	fullness per kcal of an average food	1.36	mean of Holt's 38 foods
$P^{\max}$	protein the liver can handle	3.6 g/kg × 63.3 kg = 228 g/(p·d)	Bilsborough and Mann 2006; EFSA mean weight
$Q^{\mathcal{F}}$	all fruit	4,518 g/(p·d)	one watermelon (USDA portion; M86)
r_i	daily need of component i	per nutrient	EFSA 2017; Japanese DRI 2025 (sodium)
W^{need}	total water	2,250 ml/(p·d)	EFSA 2010
m	unit of diminishing returns in well-being	100 kcal	definition

4. Inputs and variables

Symbol	Meaning	Domain	Unit
N	people of the city	$N > 0$	persons
S_g	stock available this year	$S_g \geq 0$	t
w_k	share of people in work kind k	$\sum_k w_k = 1$	–
q_g	**decision:** grams eaten per person a day	$q_g \geq 0$	g/(p·d)

5. Definitions

Availability:

$$a_g = \frac{10^6 \, S_g}{365 \, N} \tag{1}$$

Stimulants (stage 0) are drunk while available, up to the dose γ:

$$q_g = \min\Big(a_g, \; \frac{\gamma - \text{caffeine already drunk}}{\ell_g \chi_g}\Big) \quad (g \in \mathcal{T}), \qquad c = \frac{\sum_{g \in \mathcal{T}} \ell_g \chi_g q_g}{\gamma} \tag{2}$$

Need and ceilings:

$$E^{\text{need}} = B \sum_k w_k \Lambda_k + \kappa c, \qquad E^{\max} = B \Lambda^{\max} + \kappa c \tag{3}$$

$$r_{\text{energy}} = E^{\text{need}}, \qquad r_{\text{fullness}} = \bar{\sigma} E^{\text{need}}, \qquad F^{\max} = \bar{\sigma} E^{\max} \tag{4}$$

What a diet q delivers:

$$E(\mathbf{q}) = \sum_g e_g q_g, \quad F(\mathbf{q}) = \sum_g \sigma_g e_g q_g, \quad P(\mathbf{q}) = \sum_g p_g q_g, \quad P^{\mathrm{d}}(\mathbf{q}) = \sum_g d_g p_g q_g \tag{5}$$

$$c_i(\mathbf{q}) = \frac{\text{amount of } i \text{ delivered}}{r_i}: \quad c_{\text{fullness}} = \frac{F}{r_{\text{fullness}}},\ c_{\text{energy}} = \frac{E}{r_{\text{energy}}},\ c_{\text{protein}} = \frac{P^{\mathrm{d}}}{r_{\text{protein}}},\ c_i = \frac{\sum_g n_{gi} q_g}{r_i} \tag{6}$$

Nutrition adequacy: the mean adequacy ratio. Each component counts up to its need, so a surplus of one cannot make up a lack of another:

$$A(\mathbf{q}) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \min\bigl(1,\ c_i(\mathbf{q})\bigr) \tag{7}$$

Well-being: diminishing returns per 100 kcal of each good, so variety pleases more than repetition:

$$\Psi(\mathbf{q}) = \sum_{g \in \mathcal{W}} \ln\!\left(1 + \frac{e_g q_g}{m}\right) \tag{8}$$

6. The program

Feasible diets:

$$\mathcal{Q} = \left\{\mathbf{q}:\ 0 \le q_g \le \min(a_g, u_g);\ \ F(\mathbf{q}) \le F^{\max};\ \ E(\mathbf{q}) \le E^{\max};\ \ P(\mathbf{q}) \le P^{\max};\ \ \sum_{g \in \mathcal{F}} q_g \le Q^{\mathcal{F}}\right\} \tag{9}$$

Three stages, each keeping the optimum of the one before (lexicographic optimisation):

$$\text{Stage 1 (nutrition):} \quad A^* = \max_{\mathbf{q} \in \mathcal{Q}} A(\mathbf{q}) \tag{10}$$

$$\text{Stage 2 (well-being):} \quad \Psi^* = \max_{\mathbf{q} \in \mathcal{Q}}\ \Psi(\mathbf{q})\ \text{ s.t. }\ A(\mathbf{q}) \ge A^* \tag{11}$$

$$\text{Stage 3 (even use of supply):} \quad \mathbf{q}^* = \arg\min_{\mathbf{q} \in \mathcal{Q}} \sum_g \frac{q_g}{a_g}\ \text{ s.t. }\ A \ge A^*,\ \Psi \ge \Psi^* \tag{12}$$

- **Stage 3** decides only among diets equal in nutrition and well-being: it takes from the more plentiful stocks.
- **Each stage is a linear program.** In (7), $\min(1, c_i)$ becomes auxiliary variables $y_i \le 1$, $y_i \le c_i$. In (8), $\ln(1 + \cdot)$ becomes a concave piecewise-linear function with breakpoints at 0.5, 1, 2, 3, 5, 8, 12, 20 and 30 units of 100 kcal.
- **The game** solves the same three small programs, about 25 goods and 20 constraints per city per yearly step, with a bounded simplex method. The reference implementation uses HiGHS.

7. Outcomes

$$x = \frac{E(\mathbf{q}^*)}{E^{\text{need}}}, \qquad \mathit{fed} = \frac{F(\mathbf{q}^*)}{r_{\text{fullness}}}, \qquad c_i = c_i(\mathbf{q}^*), \qquad A^*,\ \Psi^* \tag{13}$$

$$\mathit{harm}_{\text{alcohol}} = \sum_{g \in \mathcal{H}} \alpha_g q_g^*, \qquad \mathit{harm}_{\text{tobacco}} = \alpha_{\text{tob}} q_{\text{tob}}^* \tag{14}$$

$$W^{\text{in}} = \sum_{g \notin \mathcal{D}} \omega_g q_g^* + \sum_{g \in \mathcal{D}} h_g q_g^*, \qquad W^{\text{plain}} = \max\bigl(0,\ W^{\text{need}} - W^{\text{in}}\bigr) \tag{15}$$

$$work = \min(1,\ x) \tag{16}$$

$$U_g = \frac{365\, N\, q_g^*}{10^6}, \qquad L_g = S_g - U_g, \qquad D_g = \delta_g L_g \tag{17}$$

$$\lambda_g^{(1)} = \frac{\partial A^*}{\partial a_g}, \qquad \lambda_g^{(2)} = \frac{\partial \Psi^*}{\partial a_g} \tag{18}$$

- **(13)** x (energy sufficiency) feeds the growth formula, which also holds body reserves. *fed,* the coverage c_i and A^* feed fertility and the deficiency burden. Ψ^* feeds the happiness line.
- **(14)** feeds the health burden. Its harm coefficients are still to be sourced.
- **(15)** In the hydration equation, h_g counts each ml of a drink as h_g ml of water: 1 L of milk gives 1.5 L of water equivalent. Plain water comes from the city's supply (D31).
- **(16)** A shortfall of energy lowers work. Surplus energy goes to activity and to the growth formula, not to extra output.
- **(18)** These are the shadow values of the stocks: what one more gram of stock would add to adequacy, or to well-being. They are zero for goods in surplus. They are the basis for prices (section 8).

8. Prices (interface, to be established)

The model gives each good two scarcity values per city, $\lambda_g^{(1)}$ and $\lambda_g^{(2)}$, in adequacy and well-being per gram.

- A good that is plentiful has value 0. A good whose stock limits nutrition or well-being has a positive value.
- These are the quantities a price must reflect for the city's population as a whole.
- How they become gold prices in the game (a scale per currency, costs of production, trade between cities) belongs to the goods economy and is not set here.

9. Change log

Version	Change
1-3 (M85-M98)	the earlier models; history in `round_m85.md` to `round_m96.md`, `food_model_assessment.md`
4 (M101)	**New model.**

Removed in version 4:

- the satisfaction factor s_g and its constants;
- the nutrient score ν_g and E^{ref};
- the well-being formula β_g and its five weights and lightness constant;
- the keys $K^{(1)}, K^{(2)}$, the two passes and the greedy algorithm;
- servings m_g per good;
- the work bonus Ω and the activity floor in work;
- the iron absorption line (a_0, a_1);

- the salt ceiling;
- the protein ceiling tied to $E^{\max}$.

Added in version 4:

- the mean adequacy ratio A (7) with fullness and energy as components;
- well-being Ψ (8) per 100 kcal;
- the staples set $\mathcal{S}$ and the well-being set $\mathcal{W}$;
- the lexicographic program (10)-(12), solved exactly;
- bioavailability b_{gi};
- the liver ceiling $P^{\max}$;
- harm outcomes (14);
- shadow values (18).

Corrected in version 4:

- F15 of the assessment is withdrawn: the hydration index is hydration relative to the same volume of water, as used in (15).
- Body reserves, seasons and classes are not part of this model (P6).